Reinforcement Learning over ψ and normalized equity

We model the control problem as a Markov decision process (MDP)

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma).$$

At each time step t, the state is

$$s_t = \left(C_t^{\max}, C_t^{\min}, C_t^{\text{avg}}, r_t^{\max}, r_t^{\min}, r_t^{\text{avg}}\right).$$

The policy chooses a continuous control action

$$a_t \sim \pi_\theta(a_t \mid s_t), \qquad a_t = \psi_t \in [0, 1].$$

The endogenous lender power for bank i is computed from its equity

$$\psi_t^i = \frac{E_t^i}{E_t^{i,\max}}.$$

The interest rate for borrower j with lender i is

$$r_t^{i,j} = \frac{m}{p_t^j(1 - \psi_t^i)}.$$

The reward is the system normalized equity

$$\mathcal{R}_t = \sum_i \frac{E_t^i}{\max_k(E_t^k)}.$$

The objective is to maximize the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t \mathcal{R}_t\right].$$

PPO optimizes the clipped surrogate objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\,\hat{A}_t, \text{clip}(\rho_t(\theta), 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t\right)\right],$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}.$$

The advantage estimate is computed with GAE:

$$\delta_t = \mathcal{R}_t + \gamma V(s_{t+1}) - V(s_t), \qquad \hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}.$$